

How to configure AD FS Toking Certificate

- 1- Open a Windows PowerShell command window and execute the following command to list the certificate of the AD FS: **Get-AdfsProperties | select *certificate***

Note: if run this command will see more information for the Certificate **Get-AdfsProperties**

Once you run this command will see the below screenshot, copy the out put and pes in the note file.

```
PS C:\Windows\system32> Get-AdfsProperties | select *certificate*

AutoCertificateRollover      : True
CertificateCriticalThreshold : 2
CertificateDuration          : 365
CertificateGenerationThreshold : 20
CertificatePromotionThreshold : 5
CertificateRolloverInterval  : 720
CertificateSharingContainer   : CN=f6a93bb9-31e6-4393-8821-d6d9d6dc176c,CN=ADFS,CN=Microsoft,CN=Program
                              Data,DC=tatimah,DC=local
CertificateThresholdMultiplier : 1440
```

- 2- Will change the Certificate for **Duration**, **Generation**, and **Promotion** form defeated to 1095 Duration (**3 years**), 30 for Generate 25 for Promotion.
 - a, C:\Set-AdfsProperties -CertificateDuration 1095
 - b, C:\Set-AdfsProperties -CertificateGenerationThreshold 30
 - c, C:\Set-AdfsProperties -CertificatePromotionThreshold 25

```
PS C:\Windows\system32> Get-AdfsProperties | select *certificate*

AutoCertificateRollover      : True
CertificateCriticalThreshold : 2
CertificateDuration          : 365
CertificateGenerationThreshold : 20
CertificatePromotionThreshold : 5
CertificateRolloverInterval  : 720
CertificateSharingContainer   : CN=f6a93bb9-31e6-4393-8821-d6d9d6dc176c,CN=ADFS,CN=Microsoft,CN=Program
                              Data,DC=tatimah,DC=local
CertificateThresholdMultiplier : 1440

PS C:\Windows\system32> Set-AdfsProperties -CertificateDuration 1095
PS C:\Windows\system32> Set-AdfsProperties -CertificateGenerationThreshold 30
PS C:\Windows\system32> Set-AdfsProperties -CertificatePromotionThreshold 25
PS C:\Windows\system32> Get-AdfsProperties | select *certificate*

AutoCertificateRollover      : True
CertificateCriticalThreshold : 2
CertificateDuration          : 1095
CertificateGenerationThreshold : 30
CertificatePromotionThreshold : 25
CertificateRolloverInterval  : 720
CertificateSharingContainer   : CN=f6a93bb9-31e6-4393-8821-d6d9d6dc176c,CN=ADFS,CN=Microsoft,CN=Program
                              Data,DC=tatimah,DC=local
CertificateThresholdMultiplier : 1440
```

- 3- This step will generate a new Certificate with the new setting and will come as a scoundrel certificate, as per the screenshots.

- a, C:\Update-ADFSCertificate -CertificateType Token-Decrypting.
- b, C:\Update-ADFSCertificate -CertificateType token-signing.

```
PS C:\Windows\system32> Update-ADFSCertificate -CertificateType Token-Decrypting
PS C:\Windows\system32> Update-ADFSCertificate -CertificateType token-signing
```

The screenshot shows the AD FS console with the 'Certificates' list. The 'Token-signing' certificate is highlighted, and the 'View Certificate...' option is selected in the Actions pane. The certificate details are as follows:

Subject	Issuer	Effective Date	Expiration Date	Status	Primary
Service communications					
CN=tatimahcapital.sa, O=شركة العالمية لـ, L=Riyadh, S=Riyadh Province, C=SA	CN=DigiCert Global G2 TL...	10/31/2023	10/17/2024		
Token-decrypting					
CN=ADFS Encryption - adfs.tatimahcapital.sa	CN=ADFS Encryption - ad...	10/31/2023	10/30/2024	Primary	
CN=ADFS Encryption - adfs.tatimahcapital.sa	CN=ADFS Encryption - ad...	11/5/2023	11/4/2026	Secondary	
Token-signing					
CN=ADFS Signing - adfs.tatimahcapital.sa	CN=ADFS Signing - adfs...	10/31/2023	10/30/2024	Primary	
CN=ADFS Signing - adfs.tatimahcapital.sa	CN=ADFS Signing - adfs...	11/5/2023	11/4/2026	Secondary	

- 4- Will promote the newly generated secondary certificate to the primary Certificate as per the below screenshot.
Note important: before promoting a secondary certificate you need to disable Auto Certificate Rollover(**false**).

C:\ Set-AdfsProperties -AutoCertificateRollover \$false

```
PS C:\Windows\system32> Set-AdfsProperties -AutoCertificateRollover $false
```

Note: now the option for set as primary will come to be activated

The image shows two screenshots of the AD FS Management console. The top screenshot shows the 'Certificates' tab with a table of certificates. The 'CN=ADFS Signing - adfs.tatimahcapital.sa' certificate is highlighted, and the 'Set as Primary' option is visible in the right-hand pane. A red box highlights the 'Set as Primary' button. The bottom screenshot shows the same console after the promotion, with the 'CN=ADFS Signing - adfs.tatimahcapital.sa' certificate now marked as 'Primary'. A red arrow points to the 'Set as Primary' button in the right-hand pane. A dialog box titled 'AD FS Management' is also visible, asking 'Setting this certificate to be the primary token-signing certificate will break the trust relationship with any relying party that does not have this certificate. Do you want to continue?'. The 'Yes' button is highlighted with a red box and a red arrow.

- 5- Export this certificate and import on **All exchange server** you have, import this certificate on personal and Trusted Root.

Note: if you have any application, you will import this certificate to all applications if you have or any application configure on ADFS.

- 6- After importing the certificate on the Exchange Server, you will update this certificate on the origination.

A, open exchange Management Center as admin and run this command **Get-OrganizationConfig |select *adfs***

```
AdfsAuthenticationConfiguration : <service><federatedAuthentication><wsFederation passiveRedirectEnabled="true"
issuer="https://adfs.tatimahcapital.sa/adfs/ls/" realm="https://fakerealm/"
requireHttps= true /><cookieHandler requireSsl= true" path="/"
/></federatedAuthentication><certificateValidation
certificateValidationMode="PeerOrChainTrust" /><audienceUris><add
value="https://mail.tatimahcapital.sa/owa/" /><add
value="https://mail.tatimahcapital.sa/owa/" /><add
value="https://mail.tatimahcapital.sa/ecp/" /><add
value="https://mail.tatimahcapital.sa/ecp/" /></audienceUris><issuerNameRegistry
type="Microsoft.IdentityModel.Tokens.ConfigurationBasedIssuerNameRegistry,
Microsoft.IdentityModel, Version=3.5.0.0, Culture=neutral,
PublicKeyToken=31bf3856ad364e35"><trustedIssuers><add
thumbprint="7F90CED38341E9917E50EFA29FA6D120A53B642E" name="Adfs"
/></trustedIssuers></issuerNameRegistry></service>
AdfsIssuer : https://adfs.tatimahcapital.sa/adfs/ls/
AdfsAudienceUris : {https://mail.tatimahcapital.sa/owa/, https://mail.tatimahcapital.sa/owa,
https://mail.tatimahcapital.sa/ecp/, https://mail.tatimahcapital.sa/ecp}
AdfsSignCertificateThumbprints : {7F90CED38341E9917E50EFA29FA6D120A53B642E}
AdfsEncryptCertificateThumbprint :
```

B, Set-OrganizationConfig -AdfsSignCertificateThumbprint **6FAA0AC86159262A0404BE29676366B7ED39C3C6**
(this cord from new Token-Signing) and restart IIS for Exchange server.

```
[PS] C:\Windows\system32>Set-OrganizationConfig -AdfsSignCertificateThumbprint f7fe99412e48ad0eab35118b942807a0960cefa5-
WARNING: For these configuration changes to take effect, you must restart Internet Information Services (IIS). To restart IIS, run
"net stop was /y" command followed by "net start w3svc" command.
[PS] C:\Windows\system32>net stop was /y
The following services are dependent on the Windows Process Activation Service service.
Stopping the Windows Process Activation Service service will also stop these services.

World Wide Web Publishing Service
Net.Tcp Listener Adapter
Net.Pipe Listener Adapter
Net.Msmq Listener Adapter

The World Wide Web Publishing Service service is stopping.....
The World Wide Web Publishing Service service was stopped successfully.

The Net.Tcp Listener Adapter service is stopping.
The Net.Tcp Listener Adapter service was stopped successfully.

The Net.Pipe Listener Adapter service is stopping.
The Net.Pipe Listener Adapter service was stopped successfully.

The Net.Msmq Listener Adapter service is stopping.
The Net.Msmq Listener Adapter service was stopped successfully.

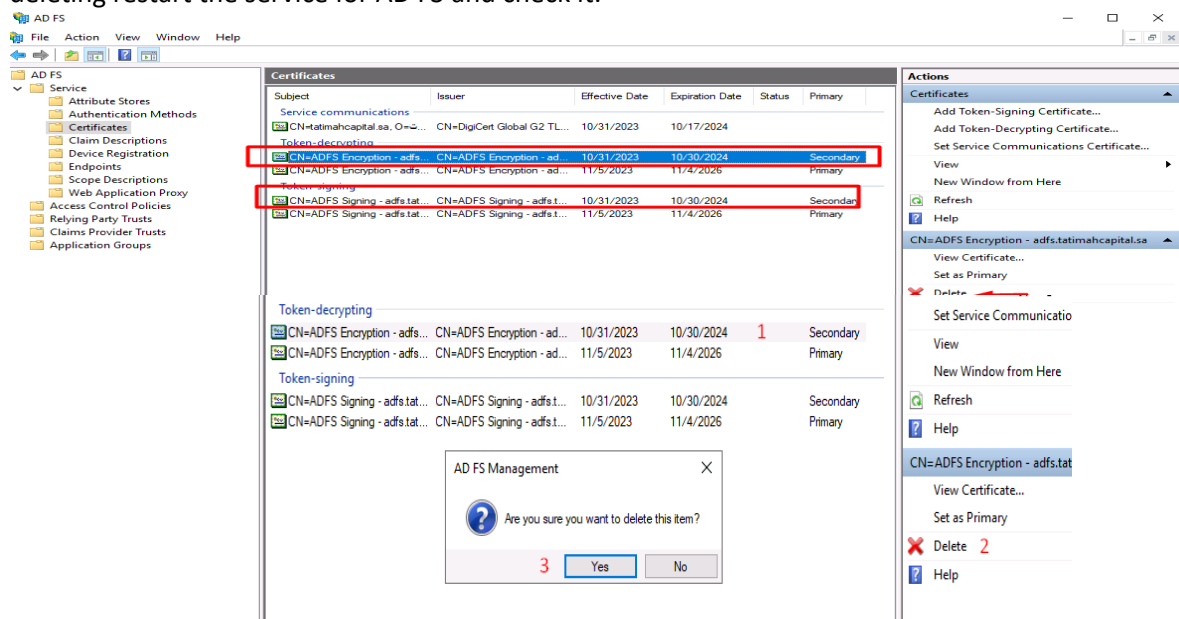
The Windows Process Activation Service service is stopping.
The Windows Process Activation Service service was stopped successfully.

[PS] C:\Windows\system32>net start w3svc
The World Wide Web Publishing Service service is starting.
The World Wide Web Publishing Service service was started successfully.

[PS] C:\Windows\system32>
```

7- Once you make the new certificate primary, you can delete the OLD certificate, by following the screenshot.

Note: after deleting restart the service for AD FS and check it.



8- Once you delete the old certificate revert Auto Certificate Rollover to enable (true).

Set-AdfsProperties -AutoCertificateRollover \$true.

```
PS C:\Windows\system32> Set-AdfsProperties -AutoCertificateRollover $true
```

9- Verification for the new Certificate, follow the below command and screenshots.

C:\ Get-AdfsProperties | select *certificate*

```
PS C:\Windows\system32> Get-AdfsProperties | select *certificate*

AutoCertificateRollover           : True
CertificateCriticalThreshold      : 2
CertificateDuration               : 1095
CertificateGenerationThreshold    : 30
CertificatePromotionThreshold     : 25
CertificateRolloverInterval       : 720
CertificateSharingContainer       : CN=f6a93bb9-31e6-4393-8821-d6d9d6dc176c,CN=ADFS,CN=Microsoft,CN=Program
                                  Data,DC=tatimah,DC=local
CertificateThresholdMultiplier    : 1440
```

C:\ Get-ADFSertificate -CertificateType Token-Decrypting

```
PS C:\Windows\system32> Get-ADFSertificate -CertificateType Token-Decrypting

Certificate      : [Subject]
                  CN=ADFS Encryption - adfs.tatimahcapital.sa
                  [Issuer]
                  CN=ADFS Encryption - adfs.tatimahcapital.sa
                  [Serial Number]
                  1DB725F7EFF9CE8342A45C35CE96B5D4
                  [Not Before]
                  11/5/2023 10:26:38 AM
                  [Not After]
                  11/4/2026 10:26:38 AM
                  [Thumbprint]
                  A7A800372665DC5E1614CD96235E307B7BAA3D35
CertificateType : Token-Decrypting
IsPrimary       : True
StoreLocation   : CurrentUser
StoreName       : My
Thumbprint      : A7A800372665DC5E1614CD96235E307B7BAA3D35
```

C:\ Get-ADFSertificate -CertificateType token-signing

```
PS C:\Windows\system32> Get-ADFSertificate -CertificateType token-signing

Certificate      : [Subject]
                  CN=ADFS Signing - adfs.tatimahcapital.sa
                  [Issuer]
                  CN=ADFS Signing - adfs.tatimahcapital.sa
                  [Serial Number]
                  2D44FF8193E051A3434537E589698F78
                  [Not Before]
                  11/5/2023 10:27:21 AM
                  [Not After]
                  11/4/2026 10:27:21 AM
                  [Thumbprint]
                  7F90CED38341E9917E50EFA29FA6D120A53B642E
CertificateType : Token-Signing
IsPrimary       : True
StoreLocation   : CurrentUser
StoreName       : My
Thumbprint      : 7F90CED38341E9917E50EFA29FA6D120A53B642E
```

Troubleshooting

If you see the below screenshot for Error update the AD FS certificate, will disable Auto Certificate Rollover, follow the below command and screenshots.

- 1- Set-AdfsProperties -AutoCertificateRollover \$true
- 2- Update-ADFSertificate -CertificateType Token-Decrypting
- 3- Update-ADFSertificate -CertificateType token-signing
- 4- Set-AdfsProperties -AutoCertificateRollover \$false
- 5- Set-AdfsProperties -AutoCertificateRollover \$true

```
PS C:\Windows\system32> Update-ADFSertificate -CertificateType Token-Decrypting
Update-ADFSertificate : PS0146: Certificate Rollover is disabled. You cannot do this operation under this mode
.
At line:1 char:1
+ Update-ADFSertificate -CertificateType Token-Decrypting
+ ~~~~~
+ CategoryInfo          : InvalidOperation: (:) [Update-AdfsCertificate], InvalidOperationException
+ FullyQualifiedErrorId : PS0146,Microsoft.IdentityServer.Management.Commands.UpdateCertificateCommand

PS C:\Windows\system32> Set-AdfsProperties -AutoCertificateRollover $true
```

```
PS C:\Windows\system32> Set-AdfsProperties -AutoCertificateRollover $true
PS C:\Windows\system32> Update-ADFSertificate -CertificateType Token-Decrypting
PS C:\Windows\system32> Update-ADFSertificate -CertificateType token-signing
PS C:\Windows\system32> Set-AdfsProperties -AutoCertificateRollover $false
PS C:\Windows\system32> Set-AdfsProperties -AutoCertificateRollover $true
```

- If you change the URL for the AD FS follow the below screenshot.

Set-OrganizationConfig -AdfsIssuer "https://adfs.atmc.com.sa/adfs/ls/"

```
[PS] C:\Scripts>Get-OrganizationConfig | select *adfs*

AdfsAuthenticationConfiguration : <service><federatedAuthentication><wsFederation passiveRedirectEnabled="true"
                                issuer="https://adfs.atmc.com.sa/" realm="https://fakerealm/" requireHttps="true"
                                /><cookieHandler requireSsl="true" path="/" /></federatedAuthentication><certificateValidation
                                certificateValidationMode="PeerOrChainTrust" /><audienceUris><add
                                value="https://mail.atmc.com.sa/owa/" /><add value="https://mail.atmc.com.sa/owa/" /><add
                                value="https://mail.atmc.com.sa/ecp/" /><add value="https://mail.atmc.com.sa/ecp/"
                                /></audienceUris><issuerNameRegistry
                                type="Microsoft.IdentityModel.Tokens.ConfigurationBasedIssuerNameRegistry,
                                Microsoft.IdentityModel, Version=3.5.0.0, Culture=neutral,
                                PublicKeyToken=31bf3856ad364e35"><trustedIssuers><add
                                thumbprint="ef6a2e9f70847a023be45af76da515f00e8601d4" name="Adfs"
                                /></trustedIssuers></issuerNameRegistry></service>

AdfsIssuer                      : https://adfs.atmc.com.sa/
AdfsAudienceUris                : {https://mail.atmc.com.sa/owa/, https://mail.atmc.com.sa/owa, https://mail.atmc.com.sa/ecp/,
                                https://mail.atmc.com.sa/ecp}
AdfsSignCertificateThumbprints  : {ef6a2e9f70847a023be45af76da515f00e8601d4}
AdfsEncryptCertificateThumbprint :

[PS] C:\Scripts>Set-OrganizationConfig -AdfsIssuer "https://adfs.atmc.com.sa/adfs/ls/"
WARNING: Commands that are available in the new remote session are different than those available when the implicit remoting module
was created. Consider creating the module again by using the Export-PSsession cmdlet.
WARNING: For these configuration changes to take effect, you must restart Internet Information Services (IIS). To restart IIS, run
"net stop was /y" command followed by "net start w3svc" command.
[PS] C:\Scripts>Get-OrganizationConfig | select *adfs*

AdfsAuthenticationConfiguration : <service><federatedAuthentication><wsFederation passiveRedirectEnabled="true"
                                issuer="https://adfs.atmc.com.sa/adfs/ls/" realm="https://fakerealm/" requireHttps="true"
                                /><cookieHandler requireSsl="true" path="/" /></federatedAuthentication><certificateValidation
                                certificateValidationMode="PeerOrChainTrust" /><audienceUris><add
                                value="https://mail.atmc.com.sa/owa/" /><add value="https://mail.atmc.com.sa/owa/" /><add
                                value="https://mail.atmc.com.sa/ecp/" /><add value="https://mail.atmc.com.sa/ecp/"
                                /></audienceUris><issuerNameRegistry
                                type="Microsoft.IdentityModel.Tokens.ConfigurationBasedIssuerNameRegistry,
                                Microsoft.IdentityModel, Version=3.5.0.0, Culture=neutral,
                                PublicKeyToken=31bf3856ad364e35"><trustedIssuers><add
                                thumbprint="ef6a2e9f70847a023be45af76da515f00e8601d4" name="Adfs"
                                /></trustedIssuers></issuerNameRegistry></service>

AdfsIssuer                      : https://adfs.atmc.com.sa/adfs/ls/
AdfsAudienceUris                : {https://mail.atmc.com.sa/owa/, https://mail.atmc.com.sa/owa, https://mail.atmc.com.sa/ecp/,
                                https://mail.atmc.com.sa/ecp}
AdfsSignCertificateThumbprints  : {ef6a2e9f70847a023be45af76da515f00e8601d4}
AdfsEncryptCertificateThumbprint :
```